Homework 6, Question 4a Verification

Solution Check and Explanation

1 Problem Statement

Let $\{N_t\}$ be a Poisson process with rate λ.

Compute: $P(N_4 = 5 \mid N_9 = 13)$

2 Your Solution: Step-by-Step Verification

VERDICT: YOUR WORK IS CORRECT!

Your approach is exactly right. You properly used:

- Bayes' rule / definition of conditional probability
- Independence of increments (crucial!)
- Poisson PMF formulas
- Algebraic simplification

This is the standard, rigorous way to solve this problem.

2.1 Step 1: Apply Definition of Conditional Probability

You wrote:

$$P(N_4 = 5 \mid N_9 = 13) = \frac{P(N_4 = 5, N_9 = 13)}{P(N_9 = 13)}$$

Why this is correct: This is the definition of conditional probability.

CORRECT

2.2 Step 2: Rewrite Joint Probability Using Increments

You wrote:

$$P(N_4 = 5, N_9 = 13) = P(N_4 = 5, N_9 - N_4 = 8)$$

Why this is correct: If $N_4 = 5$ and $N_9 = 13$, then the increment $N_9 - N_4 = 13 - 5 = 8$. This is just rewriting the same event.

CORRECT

2.3 Step 3: Use Independence of Increments

You wrote:

$$P(N_4 = 5, N_9 - N_4 = 8) = P(N_4 = 5) \cdot P(N_9 - N_4 = 8)$$

Why this is correct: This uses the **independence of increments** property of Poisson processes:

> **Independence of Increments**
>
> For a Poisson process, non-overlapping increments are independent.
> Specifically: N_4 (events in $[0, 4]$) and $N_9 - N_4$ (events in $(4, 9]$) are independent random variables.

CORRECT - This is the key insight!

2.4 Step 4: Apply Poisson PMF Formulas

You wrote:

$$P(N_4 = 5) = \frac{e^{-4\lambda}(4\lambda)^5}{5!}$$
$$P(N_9 - N_4 = 8) = \frac{e^{-5\lambda}(5\lambda)^8}{8!}$$
$$P(N_9 = 13) = \frac{e^{-9\lambda}(9\lambda)^{13}}{13!}$$

Why this is correct:

- $N_4 \sim \text{Poisson}(4\lambda)$ (events in time interval of length 4)
- $N_9 - N_4 \sim \text{Poisson}(5\lambda)$ (events in time interval of length $9 - 4 = 5$)
- $N_9 \sim \text{Poisson}(9\lambda)$ (events in time interval of length 9)

Each uses the standard Poisson PMF: $P(X = k) = \frac{e^{-\mu}\mu^k}{k!}$
CORRECT

2.5 Step 5: Substitute Into Conditional Probability

You wrote:

$$P(N_4 = 5 \mid N_9 = 13) = \frac{\frac{e^{-4\lambda}(4\lambda)^5}{5!} \cdot \frac{e^{-5\lambda}(5\lambda)^8}{8!}}{\frac{e^{-9\lambda}(9\lambda)^{13}}{13!}}$$

Why this is correct: Direct substitution of the formulas from Steps 1, 3, and 4.
CORRECT

2.6 Step 6: Simplify

You wrote:

$$= \frac{e^{-4\lambda} \cdot e^{-5\lambda}}{e^{-9\lambda}} \cdot \frac{(4\lambda)^5 \cdot (5\lambda)^8}{(9\lambda)^{13}} \cdot \frac{13!}{5! \cdot 8!}$$

Then:

$$= \frac{e^{-9\lambda}}{e^{-9\lambda}} \cdot \frac{\lambda^5 \cdot \lambda^8}{\lambda^{13}} \cdot \frac{4^5 \cdot 5^8}{9^{13}} \cdot \frac{13!}{5! \cdot 8!}$$

Why this is correct:

- Exponentials: $e^{-4\lambda} \cdot e^{-5\lambda} = e^{-9\lambda}$ (cancels with denominator)
- Powers of λ: $\lambda^5 \cdot \lambda^8 = \lambda^{13}$ (cancels with denominator)
- Factorials: $\frac{13!}{5! \cdot 8!} = \binom{13}{5}$

CORRECT

2.7 Step 7: Final Form

You wrote:

$$= \binom{13}{5} \left(\frac{4}{9}\right)^5 \left(\frac{5}{9}\right)^8$$

Why this is correct: After all cancellations:

$$\frac{4^5 \cdot 5^8}{9^{13}} = \frac{4^5}{9^5} \cdot \frac{5^8}{9^8} = \left(\frac{4}{9}\right)^5 \left(\frac{5}{9}\right)^8$$

CORRECT

2.8 Step 8: Numerical Calculation

You calculated:

$$\binom{13}{5} \left(\frac{4}{9}\right)^5 \left(\frac{5}{9}\right)^8 \approx 0.203$$

Verification:

- $\binom{13}{5} = 1287$
- $(4/9)^5 \approx 0.01734$
- $(5/9)^8 \approx 0.00907$
- Product: $1287 \times 0.01734 \times 0.00907 \approx 0.2025$

CORRECT (0.203 or 20.3%)

3 Why Your Method Works: The Big Picture

Your solution brilliantly demonstrates a fundamental strategy for Poisson process problems:

Problem-Solving Strategy

When finding conditional probabilities for Poisson processes:

1. Use Bayes' rule: $P(A \mid B) = \frac{P(A,B)}{P(B)}$
2. Rewrite joint events using **increments**
3. Apply **independence of increments** to split the joint probability
4. Use Poisson PMF formulas for each term
5. Simplify (often λ cancels out!)

3.1 Why λ Cancels Out

Notice something remarkable: **the final answer doesn't depend on λ!**

This makes intuitive sense:

- We're asking: "Given 13 arrivals by time 9, what's the probability that 5 of them arrived by time 4?"
- The *rate* doesn't matter—only the *proportion* of time: $\frac{4}{9}$ of the time has passed
- Each of the 13 arrivals is "equally likely" to have arrived in $[0, 4]$ vs $(4, 9]$

4 The Direct Formula (Shortcut)

Your derivation proves this general result:

Conditional Distribution Formula

For a Poisson process $\{N_t\}$ with rate λ, and $0 < t < s$:

$$\boxed{(N_t \mid N_s = n) \sim \text{Binomial}\left(n, \frac{t}{s}\right)}$$

In other words:

$$P(N_t = k \mid N_s = n) = \binom{n}{k} \left(\frac{t}{s}\right)^k \left(1 - \frac{t}{s}\right)^{n-k}$$

For this problem: $t = 4$, $s = 9$, $n = 13$, $k = 5$:

$$P(N_4 = 5 \mid N_9 = 13) = \binom{13}{5} \left(\frac{4}{9}\right)^5 \left(\frac{5}{9}\right)^8$$

Your work derived this formula from first principles!

5 Intuition: Why Binomial?

Think of it this way:

"Given 13 arrivals happened by time 9, what's the probability that exactly 5 of them arrived by time 4?"

Each of the 13 arrivals has probability $\frac{4}{9}$ of arriving in $[0, 4]$ (since time 4 is $\frac{4}{9}$ of the way to time 9).

So we're doing 13 independent trials (one for each arrival), each with success probability $\frac{4}{9}$, and asking for exactly 5 successes.

That's exactly a **Binomial(13, 4/9)** distribution!

6 Common Mistakes (You Avoided These!)

6.1 Mistake 1: Forgetting Independence

Wrong: Not recognizing that N_4 and $N_9 - N_4$ are independent

You did it right: You explicitly used independence to split the joint probability

6.2 Mistake 2: Incorrect Increment Distribution

Wrong: Thinking $(N_9 - N_4) \sim \text{Poisson}(9\lambda - 4\lambda) = \text{Poisson}(5\lambda)$ by subtracting parameters

You did it right: You correctly identified that the increment over an interval of length 5 has parameter 5λ

6.3 Mistake 3: Arithmetic Errors

Wrong: Messing up the simplification (forgetting to cancel λ, etc.)

You did it right: Your algebra is clean and all cancellations are correct

7 Summary and Grade

Final Assessment

Is your work correct? YES!
Is this the right process? YES!
Is this the right answer? YES!

Grade: 10/10
Your solution demonstrates:

- Deep understanding of Poisson process properties
- Correct application of independence of increments
- Clean algebraic manipulation
- Correct final answer (≈ 0.203 or 20.3%)

This is exactly how the problem should be solved. Excellent work!

8 Key Formulas for Reference

8.1 Poisson Process Basics

- $N_t \sim \text{Poisson}(\lambda t)$ for a Poisson process with rate λ
- PMF: $P(N_t = k) = \frac{e^{-\lambda t}(\lambda t)^k}{k!}$
- Mean: $\mathbb{E}(N_t) = \lambda t$
- Variance: $\text{Var}(N_t) = \lambda t$

8.2 Independence of Increments

For $0 \leq s < t < u$:

- $N_t - N_s$ and $N_u - N_t$ are independent
- $N_t - N_s \sim \text{Poisson}(\lambda(t - s))$

8.3 Conditional Distribution

For $0 < t < s$:

$$(N_t \mid N_s = n) \sim \text{Binomial}\left(n, \frac{t}{s}\right)$$